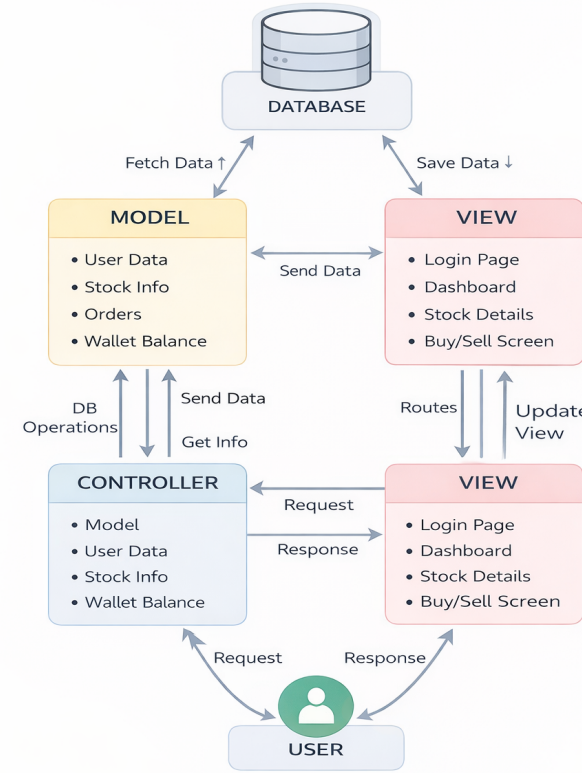
MVC PATTERN
(Model-View Controller)



Model–View–Controller (MVC) is a software architectural pattern used to organize an application into three separate components: Model, View, and Controller. This pattern helps separate the user interface, application logic, and data management, making the system easier to develop, maintain, and scale. In a Stock Trading Application, MVC ensures that each part of the application performs a specific role while still working together efficiently.

1. Model

The **Model** represents the data layer of the application. It is responsible for managing and storing data related to users, stock information, trading orders, wallet balance, and portfolio details. The model interacts directly with the database to store new transactions and retrieve information whenever required by the system.

Data stored in Model:

- User information
- Stock details
- Trading orders
- Portfolio data

The Model communicates directly with the database to store and retrieve information.

2. View

The **View** represents the user interface of the application. It displays information to the user and allows interaction with the system. Examples of views in a stock trading application include the login page, dashboard, stock details page, buy/sell screen, and portfolio page. The view receives data from the controller and presents it in a clear and user-friendly format.

Views in the application:

- Login page
- Dashboard page
- Stock details page
- Buy/Sell screen
- Portfolio page

3. Controller

The **Controller** acts as the intermediary between the Model and the View. It receives user inputs from the interface, processes the request, and communicates with the model to retrieve or update data. After processing the request, the controller sends the result back to the view so that the user interface can be updated with the latest information.

Responsibilities of Controller:

- Receive user requests
- Validate data
- Interact with Model
- Send responses to the View

MVC Workflow

1. User clicks the **Buy Stock** button.
2. The request is sent to the **Controller**.
3. Controller validates the request and sends it to the **Model**.
4. The Model updates the database with the new order.
5. Controller returns the response to the **View**.
6. The View displays the updated portfolio to the user.